

Technology Requirements

Date September 2026
Team ID xxxxxx
Project Name EduMind AI - Smart AI-Powered Learning Assistant
Maximum Marks 3 Marks

Step 1: Software Requirements

Category	Technology / Tool	Purpose
Programming Language	Python	Backend development and AI integration.
Web Framework	FastAPI	API routing and web application backend.
Frontend	HTML / CSS / JavaScript	User interface and browser interaction.
AI Service	Google Gemini API	Generative AI responses.
Python SDK	Google Gen AI SDK	Connect Python backend to Gemini.
Environment Management	python-dotenv / .env	Load API credentials without hard-coding them.
Server	Uvicorn	Run the FastAPI application locally.
IDE	Visual Studio Code	Development and project management.

Step 2: Hardware / Execution Requirements

Requirement	Details
Computer	Windows-based laptop or desktop capable of running Python and a browser.
Internet	Required for Gemini API requests and web access.
Browser	Modern browser such as Google Chrome or Microsoft Edge.
Python Environment	Python 3.x with project dependencies installed.

Step 3: Security Requirement

The Gemini API key should remain in the local environment file and must not be included in the project source, screenshots or public submission repository.